releases. The model also includes an intrusion scenario using a well drilling probability, beginning after 300 years (in the main case), as it is expected that control will last that long. It is assumed that as soon as a drill core from a waste repository is taken, it will immediately be recognized; this means that high doses from the core will only be received by drilling and laboratory personal. The risk[1] from intrusion after 100 years is calculated to be 8.1×10^{-7} y^{-1} and after 100 000 years 2.1×10^{-9} y^{-1}. By comparison, the calculated risk from the "normal" scenario of releases from spent fuel elements and migration in the environment over a long period is 5×10^{-5} y^{-1}. With this exception all the maximum calculated individual risks from disposal of solid wastes are below the limit of 10^{-5} y^{-1} recommended by the ICRP. For vitrified high level waste and spent fuel disposal, the doses calculated for intrusion scenarios are very high but the probability of occurrence are low, so the risks from intrusion are lower than those from radionuclide migration with groundwater. The maximum individual doses from the migration scenarios for solid waste disposal are predicted to arise beyond 10 000 years after closure of the repository.

These values indicate that release and migration is the more significant concern, although even for this scenario, geologic disposal is calculated to present very small risks to future generations.

22.11. CONCLUSIONS

A rather large amount of nuclear fission products and actinide elements have been released to the environment from nuclear weapons testing and from accidental and intentional discharges from nuclear reactor operations and fuel reprocessing. The research on the fate of these released radionuclides suggest that the long lived actinides form quite insoluble or strongly sorbed species while ^{129}I and ^{99}Tc have relatively high dissemination in natural systems. The most active shorter-lived species (^{90}Sr, ^{137}Cs) also have more mobility in ecosystems.

These conclusions are confirmed by the results of the investigations of radionuclide behavior in natural analog sites. Even more relevant are the results of the natural reactor region at Oklo.

The data from field studies is largely confirmed by the performance assessments of proposed nuclear waste repositories. The "intrusion" scenario is calculated to have a lower risk than the undisturbed natural leach and migration scenario. The latter qualitatively agrees with the Oklo data and indicates no unacceptable risks result from a carefully chosen and designed geologic repository in which the nuclear wastes are emplaced with appropriate packaging.

[1] Risk is here assumed to be equal to expected frequency regardless of the severity of the consequence.

22.12. EXERCISES

22.1. What is the purpose of the clay buffer of a waste repository in granitic rock?

22.2. Which of the following ions move slower than the groundwater: K^+, Cs^+, La^{3+}, TcO_4^-, HCO_3^-?

22.3. Why is Np assumed to move faster than Pu in most groundwaters?

22.4. (a) What types of geologic formations are considered for waste repositories? (b) Which types were used/planned for Asse and WIPP?

22.5. What will be the concentration of $Am(CO_3)_2^-$ at pH 7 under the conditions given in §22.7.

22.6. What will be the ratio of [Pu(III)]/Pu[(IV)] in a groundwater containing small concentrations of iron in the relation Fe(II) 99% and Fe(III) 1%? E^0 for Fe(II)/Fe(III) = 0.743 V. Neglect hydrolysis.

22.7. (a) Calculate the Pourbaix-line for Np(IV)/Np(V) in Figure 22.6. (b) What value must the *E*h value of water exceed at pH 6 for Np(V) to dominate? Neglect hydrolysis.

22.8. A miner has a deep well 160 km away from a waste repository. During an earthquake, the rock fractures and a groundwater stream opens between the repository and the well so that 10 Ci ^{90}Sr momentarily dissolves and moves toward the well at ν_w 160 km y^{-1}. (a) What will be the amount of ^{90}Sr (in Bq) reaching the well (assume plug flow)? (b) Will that water be harmful to the miner?

22.13. LITERATURE

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